

NEURAL BELLMAN OPERATORS

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This paper develops Neural Bellman Operators for policy evaluation and feasible improvement in controlled economies. A common continuation connects current choices to policy-specific future payoffs. We derive decision-loss bounds from centered continuation errors, a transport bound for changes in future economic primitives, and a true-objective certificate for continuous scalar decisions. These results give explicit conditions for reuse and refresh while retaining the boundary, utility-domain, and deviation requirements of the original economic applications. The capital evidence includes protected high-dimensional payoff comparisons, conditional prediction-risk assessment, and simultaneous finite-catalogue bounds. A new prospective experiment varies one to 1,024 queries and compares seven complete procedures at a common economic tolerance, charging fitting, querying, failed checks, and verification. NBO attains the scalar-interval target in every declared stream; conventional surrogates remain competitive and are not excluded from the cost frontier. The results distinguish a reusable learned continuation from a claim of uniform neural dominance or an unverified solution of the full diffusion control problem.

KEYWORDS: Dynamic programming, Neural approximation, Recursive utility, Endogenous preferences, Viscosity solutions, Dynamic games.

1. INTRODUCTION

A current economic choice changes a future that is costly to evaluate. A household chooses consumption while investing in its preferences; a firm adjusts capital against the policies of its rivals; an economist recomputes these choices under alternative charges or constraints. Neural Bellman Operators (NBO) evaluate a specified continuation policy and use a feasible actor to improve current decisions. The question is when paying for a learned continuation makes these decisions

more accurate, or less expensive at a specified economic accuracy, than directly
simulating or approximating them.

This question involves three distinct objects. The continuation equation specifies
the future being valued. The actor specifies the feasible decision actually taken. The
economic error account specifies what is known about its payoff. A small regression
loss does not, by itself, answer the economic question; neither does a short training
clock. The comparison must charge construction, all action queries, unsuccessful
accuracy checks, and the verification required to return a decision. It must also
allow competing procedures to reuse the information that their decision problems
permit.

The paper develops this evaluation–decision–payoff chain for controlled
economies. Its original differential and monotone formulations, recursive utility, en-
dogenous preferences, temporal selves, and strategic interaction remain substantive
components. Policy evaluation is always policy specific. A utility-domain restric-
tion cannot be discarded when the approximation is reused, and a best response
against fixed rivals is not a proof of equilibrium. These distinctions determine which
continuation can be shared and when it must be refreshed.

Our first contribution is an explicit connection between continuation reuse and
decision loss. A centered error, rather than an arbitrary level error, controls the
loss from an approximate maximizing action. A transport account then bounds
the additional loss when future economic primitives change. The resulting reuse
region includes bounded changes in transitions, rewards, and recursive evaluation
operators; it is not confined to tasks with an identical future. The bound makes
the relevant stability and utility-domain conditions explicit. It also explains why
prediction-risk rankings need not coincide with the payoff rankings of competing
policies.

Our second contribution is a work-to-accuracy implementation of this account.
For a finite-support instance of the nonlinear capital economy, an outward interval
calculation certifies the true objective’s derivative and a uniform strong-concavity
bound. Every returned action is assessed against the full continuous scalar action
interval, not merely a computed candidate list. A frozen sequence of fitting budgets
stops at the first successful economic check and charges every attempt. This supplies

a realized stopping cost, whereas the earlier simultaneous catalogue closure supplies a valid but retrospective accuracy comparison. Both calculations remain in the paper.

The completed design varies the number of decisions from one to 1,024 in dimensions ten and fifty. Temporary withdrawal charges, current utility weights, adjustment costs, and caps generate the decisions while the future reference policy stays fixed. Seven complete procedures include NBO, conventional quadratic and radial-basis continuation regressions, a shared task-conditioned actor, cached sample-average optimization, full-support enumeration, and a current-reward rule. Every dimension-volume cell is executed under three fixed fitting and cache streams. The integrated execution replays this already frozen design; it does not create additional independent streams by repeating a computation. The resulting economic certificates are deterministic for the stored finite economy.

NBO reaches the prescribed scalar-interval tolerance in all 36 of its complete runs. This identifies successful learned-continuation decisions, not uniform neural superiority. The non-neural surrogates and simulation-based procedures also achieve the target and remain on the measured cost frontier. The shared actor and current-only procedure retain their unsuccessful checks. Common exogenous actions and each predictor's own selected actions are both assessed against full-support continuation means; caches are regenerated across the three streams. Thus absolute risk, centered risk, and actual finite-menu decision loss can be compared without treating simulation noise as the target value or selecting the diagnostic action pair in favor of one predictor.

This evidence complements the original capital experiments rather than replacing them. In the previously unpiloted fifty-dimensional Long and Intermediate menus, the protected NBO payoff advantages over materialized Raw and direct-policy actors exceed 10^{-4} . The earlier conditional continuation-risk assessment yields eight comparisons favoring NBO, four favoring Raw, and twelve unresolved comparisons. None of its eight 64-path comparisons favors NBO. Exact closure of the original simultaneous payoff intervals makes final NBO the least-cost *certified* candidate in one of eight cells, while cheaper unresolved candidates remain. Those statements retain their original finite populations, probability allocations, and chronology; they are not restated as results of the new stopping design.

The continuous-time evidence also keeps its own economic target. Independent confirmation favors NBO over the original neural-HJB implementation in both dimensions. Against the strengthened exact-trace HJB procedure, only the low-volatility fifty-dimensional cell establishes the declared margin against both deployments. The signed Bellman assessment has positive continuous-economy gain bounds, while its signed continuation-correction intervals contain zero. Earlier unresolved NBO–Raw comparisons, the cheaper Raw clock, and nonpositive mechanism bounds are preserved. A finite scalar certificate does not remove diffusion, implementation, information, or full-control errors from these separate accounts.

The principal economic exercise in the new design is comparative statics for temporary resource charges. Its primary target is foregone transformed welfare at the chosen action. We derive the sign of the optimal charge response and translate the target into an external log-utility compensation. This is a specified counterfactual exercise in the paper’s capital economy, not an estimate of an observed tax reform. The task distribution defines the exercise and is not presented as an estimated population of policy requests. The results show when a common continuation supports these decisions and what its construction and certification cost. They do not require that a neural representation be the best approximation for every economic model.

The main article presents the common decision problem, its economic error chain, and the complete primary comparisons. The technical supplement gives full proofs, the detailed inherited capital analysis, and all computational records. The Economic Applications companion retains the complete utility, preference, temporal-self, and game models. This organization separates the central argument from its derivations without changing the paper’s subject or discarding unfavorable evidence.

1.1. *Relation to the literature*

Neural Bellman Operators build on numerical dynamic programming and function approximation. Projection and related methods exploit economic structure to approximate a Bellman equation; see [Judd \(1998\)](#). Smooth neural networks can approximate derivatives under representation hypotheses ([Hornik et al., 1990](#)).

These results distinguish representational capacity from the accuracy achieved by a finite architecture and a particular optimizer.

The differential critic is related to the neural PDE approach in [Sirignano and Spiliopoulos \(2018\)](#). Their DGM formulation includes differential and boundary conditions and treats HJB equations. The deep BSDE method of [Han et al. \(2018\)](#) also includes an HJB application. Our object combines a specified policy continuation, a feasible actor, and an economic error account, and asks when the continuation representation is useful across decision queries. An analytical maximizing action is useful when available; it is not a universal distinction between neural PDE methods. Recursive dependence on the value is a nonlinear zeroth-order term, whereas full nonlinearity of a PDE concerns its highest derivatives.

Recursive utility separates intertemporal substitution from risk aversion ([Epstein and Zin, 1989](#), [Duffie and Epstein, 1992](#)). Utility-process selection and transversality matter in its infinite-horizon formulation ([Herdegen et al., 2021](#)). Temporal selves instead require an equilibrium formulation ([Björk et al., 2016](#)). The stochastic approximation literature provides convergence results under explicit step-size, stability, and limiting-dynamics conditions ([Borkar, 2024](#)). We retain those distinctions when describing a finite neural computation.

Section 4 defines the economic object and Section S.27 specializes it to the capital economy. Section S.27.3 connects continuation error with economic gain. The computational design in Section 6 distinguishes continuation reuse, conditional payoff replication, and mechanism assessment. Section 11 states the other economic applications. General verification results and their full proofs remain in the technical supplement, beginning with Section S.1; the application companion is a current part of the paper.

1.2. Continuation approximation and the contribution of NBO

A reusable continuation is not a new principle of dynamic programming. Postdecision-state approximation separates a controlled transition from the subsequent uncertainty and is a standard device in approximate dynamic programming ([Powell, 2011](#)). Regression can estimate a continuation relevant to an economic

stopping decision, as in [Longstaff and Schwartz \(2001\)](#). Batch fitted value methods turn transition data into successive supervised regression problems ([Ernst et al., 2005](#)); approximate policy iteration likewise separates an evaluation error from an imperfect improvement step ([Scherrer et al., 2015](#)). These methods establish the appropriate comparison class for NBO. The relevant question is not whether continuation fitting is possible, but which fitted object, decision rule, and verification account are useful for the stated economic exercise.

The actor-critic distinction is also established. In [Konda and Tsitsiklis \(1999\)](#), a critic supplies information for a parameterized policy update under explicit stochastic-approximation conditions. NBO uses a scalar policy continuation and its derivative channels in a feasible economic improvement map. The scalar representation couples value and gradient predictions through one potential; a separately fitted vector field need not have this property. That structural distinction is testable but does not imply that a scalar neural network must outperform a costate surrogate, a direct actor, or a conventional scalar regression. The menu’s vector baseline and the new quadratic, radial-basis, and shared-actor baselines therefore remain in the primary comparisons.

The reuse comparison is closely related to simulation optimization. A cached sample-average objective can be optimized at many current tasks. Reusing innovations is legitimate; reusing their payoff at a different postdecision state without reevaluation is not. The new experiment gives cached simulation and regression the same finite-law information. The regressions share their value and gradient labels, and the shared actor receives the same task variables. Full-support enumeration is a particularly strong comparator when the law has only sixteen atoms. Its tractability is disclosed rather than suppressed to manufacture an advantage for approximation.

Several mathematical components are standard. Centering removes constants from action comparisons; the squared-error difference cancels common noise; shortest-path closure solves difference constraints; and a strong-concavity bound converts a first-order residual into an objective gap. For stochastic stopping, time-uniform confidence sequences provide a general route to valid repeated assessment ([Howard et al., 2021](#)). We do not claim any of these principles as a new theorem of

dynamic programming, graph theory, or probability. The contribution developed here is their explicit economic linkage: which continuation is reusable, how approximation and changes in future primitives enter decision loss, which computed action is certified, and which work is actually paid before that certificate is returned. The deterministic finite-law experiment uses no confidence sequence and spends no additional probability. Its scalar-interval guarantee and the earlier statistical menu guarantee are different applications of that linkage.

2. ECONOMIC DECISIONS AND ACCURACY TARGETS

A task θ and observed state s determine a feasible action set $A_\theta(s)$ and current reward $r_\theta(s, a)$. The postdecision state is $T_\theta(s, a)$. Write

$$Q_\theta(s, a) = r_\theta(s, a) + S_\theta\{T_\theta(s, a)\}, \quad L_\theta(s, a) = \sup_{b \in A_\theta(s)} Q_\theta(s, b) - Q_\theta(s, a). \quad (1)$$

The continuation S_θ includes future rewards under a specified future policy, conditional on the postdecision information. Current rewards are excluded. A change in the current charge alone need not change S_θ ; a change in the future policy generally does. For an observed catalogue $\{(s_i, \theta_i)\}_{i=1}^q$, our new primary target is $q^{-1} \sum_i L_{\theta_i}(s_i, a_i) \leq \varepsilon$.

The word “regret” below always refers to the displayed comparison class. The current scalar experiment takes $s \in \mathbb{R}^d$ but restricts the current withdrawal to a uniform scalar action. Neither a large state dimension nor full enumeration of its finite innovation law turns this target into an optimum over vector actions or all adapted future controls.

3. FROM A LEARNED CONTINUATION TO ECONOMIC ACCURACY

3.1. Centered approximation error and the selected decision

Fix an anchor continuation S_0 and its approximation $\hat{S}_0$. For each task, let $e = \hat{S}_0 - S_0$ and $\Delta_\theta = S_\theta - S_0$. The implemented action $\hat{a}$ satisfies

$$r_\theta(s, \hat{a}) + \hat{S}_0(T_\theta(s, \hat{a})) \geq \sup_{a \in A_\theta(s)} \{r_\theta(s, a) + \hat{S}_0(T_\theta(s, a))\} - \delta_\theta(s). \quad (2)$$

TABLE I
ECONOMIC TARGETS AND THEIR VERIFICATION DOMAINS

Target	Comparison object	Interpretation in this paper
Continuation risk	Exact conditional continuation at specified arguments	Prediction diagnostic; neither policy welfare nor control regret.
Finite-menu loss	Best action in one exogenous finite action menu	Decision diagnostic at common tasks; not a continuous-action bound.
Catalogue regret	Best of sixteen historical method–budget candidates	Earlier simultaneous payoff intervals and their exact closure.
Scalar-interval regret	True maximum over $[\cdot, \bar{a}_i]$ in the stored finite economy	New primary accuracy certificate and realized stopping target.
Diffusion control regret	Full declared adapted-control class in continuous time	Separate boundary, implementation, diffusion and information accounts.
Utility or game criterion	Policy utility, spike deviation or unilateral best response	Application-specific admissibility and verification conditions.

Here $\delta_\theta(s) \geq 0$ is an actual optimization allowance, not a training-loss tolerance. For a function on the feasible actions, write $\text{osc}_A f = \sup_A f - \inf_A f$.

PROPOSITION 1—Continuation error, transport, and decision loss: *Suppose the suprema in (1) and (2) are finite. Then*

$$L_\theta(s, \hat{a}) \leq \delta_\theta(s) + \text{osc}_{A_\theta(s)}(e \circ T_\theta) + \text{osc}_{A_\theta(s)}(\Delta_\theta \circ T_\theta). \quad (3)$$

For a finite action set $\{a_1, \dots, a_m\}$ with $m \geq 2$ and fixed positive weights π_j summing to one, define

$$R_\pi(s) = \sum_j \pi_j (e(T_\theta(s, a_j)) - \bar{e}_\pi(s))^2,$$

$$\bar{e}_\pi(s) = \sum_j \pi_j e(T_\theta(s, a_j)),$$

$$C_\pi = \max_{i \neq j} (\pi_i^{-1} + \pi_j^{-1}).$$

If $|\Delta_\theta| \leq b_\theta$ on the queried postdecision set, then

$$L_\theta(s, \hat{a}) \leq \delta_\theta(s) + \sqrt{C_\pi R_\pi(s)} + 2b_\theta. \quad (4)$$

Both inequalities hold for each fitted object and each realized cache. Expectations over training or cache randomness may subsequently be taken.

The proof is in Section S.28. An additive continuation error common to all feasible actions has no effect on the decision. This is why an absolute squared prediction error may be large even when decisions are accurate. Conversely, the risk at only the NBO/reference action pair cannot control the losses of another method choosing elsewhere. Our new diagnostic uses a common exogenous action set and, separately, the arguments selected by each predictor. Inequality (4) is a bound for the common finite set; it is not applied to the entire scalar interval using just three sampled actions.

If the true finite-menu best action has gap $g > 0$ from every competitor, then a bound strictly smaller than g implies that the implemented action is best. Without such a margin, arbitrarily small centered prediction error can change the identity of the selected action. Neither statement orders two methods' realized payoffs from their risks alone.

3.2. A nontrivial reuse region when the future changes

Let $X_{k+1}^\theta = F_{\theta,k}(X_k^\theta, Z_k)$ describe the future under the policy being evaluated. A common-innovation coupling need not make the transition maps identical. On an invariant state domain, assume

$$\begin{aligned} \|F_{0,k}(x, z) - F_{0,k}(x', z)\| &\leq L_k \|x - x'\|, & \|F_{\theta,k}(x, z) - F_{0,k}(x, z)\| &\leq \epsilon_{\theta,k}, \\ |g_{0,k}(x) - g_{0,k}(x')| &\leq H_k \|x - x'\|, & |g_{\theta,k}(x) - g_{0,k}(x)| &\leq \eta_{\theta,k}. \end{aligned} \quad (5)$$

These bounds concern future rewards and transitions, not just the current task. Include the terminal payoff as $g_{\theta,K}$ and use nonnegative reward weights ω_k . The initial postdecision state is shared.

THEOREM 2—Continuation transport and refresh: Set $d_0 = 0$ and $d_{k+1} = L_k d_k + \epsilon_{\theta,k}$. Under (5), uniformly on the common postdecision domain,

$$|S_\theta - S_0| \leq b_\theta := \sum_{k=0}^K \omega_k (\eta_{\theta,k} + H_k d_k). \quad (6)$$

Thus the anchor may be reused at a task whenever the right side of (3), or the applicable finite-menu bound (4), is at most the desired economic tolerance.

More generally, suppose a recursive evaluation on a common invariant utility domain has maps $V_{\theta,k} = \mathcal{G}_{\theta,k}(V_{\theta,k+1})$. If these maps are sup-norm Lipschitz with constants β_k on that domain, $\|\mathcal{G}_{\theta,k}(v) - \mathcal{G}_{0,k}(v)\|_\infty \leq \nu_{\theta,k}$, and $\|V_{\theta,K} - V_{0,K}\|_\infty \leq b_K$, then

$$\|V_{\theta,0} - V_{0,0}\|_\infty \leq \sum_{k=0}^{K-1} \left(\prod_{j=0}^{k-1} \beta_j \right) \nu_{\theta,k} + \left(\prod_{j=0}^{K-1} \beta_j \right) b_K. \quad (7)$$

Empty products equal one.

The theorem supplies a class of tasks with genuinely different futures. For example, uniformly bounded transition and reward perturbations of an anchor economy satisfy it on any verified invariant compact domain with bounded Lipschitz constants. A single learned continuation then supports a whole neighborhood of counterfactuals until its economic error allowance is spent. The neighborhood is conditional on the displayed stability bounds; it is not a distribution-free assertion about all future policies. When future rewards, constraints, or opponents' policies violate those bounds, the evaluated continuation must be recomputed or its transport allowance reestablished. For recursive utility the common utility domain is essential: a local Lipschitz constant cannot be extended across a singular utility boundary. For games, this comparison evaluates a specified deviation problem; it does not replace the fixed-point or best-response conditions in the Economic Applications companion.

3.3. An explicit amortization class

The preceding inequalities give a sufficient class in which a shared scalar continuation supports a cheaper declared accuracy procedure, not just a post hoc

interpretation of successful cells. Fix a finite exogenous query family with K distinct postdecision arguments and a common continuation $S(x) = \phi(x)^\top \beta$. The feature map $\phi : \mathcal{X} \rightarrow \mathbb{R}^p$ may be a frozen neural dictionary or a conventional basis. Its construction cost is charged, and it is fixed before the fitting noises below. At design points x_i , observe $Y_i = S(x_i) + \xi_i$, where the noises are conditionally independent, mean zero, and σ -sub-Gaussian. Let Φ have rows $\phi(x_i)^\top$ and assume

$$G_n = \Phi^\top \Phi \text{ is invertible, } \max_{x \text{ queried}} \phi(x)^\top G_n^{-1} \phi(x) \leq \ell/n.$$

For example, $G_n/n \succeq cI$ and $\|\phi(x)\|^2 \leq B_\phi^2$ imply $\ell = B_\phi^2/c$. These are checkable design conditions, not a claim about an arbitrary trained representation.

COROLLARY 3—A sufficient learned-continuation work advantage: Fix $\varepsilon > \delta \geq 0$ and $0 < \alpha < 1$. Least-squares continuation fitting and δ -accurate finite-menu maximization have loss at most ε on every query with probability at least $1 - \alpha/2$ if

$$n \geq \frac{8\sigma^2\ell}{(\varepsilon - \delta)^2} \log \frac{4K}{\alpha}, \quad (8)$$

provided the displayed full-rank and leverage conditions hold. Direct sample-average predictions of the same continuation have the same guarantee when they use at least $t = \lceil 8\sigma^2 \log(4K/\alpha) / (\varepsilon - \delta)^2 \rceil$ observations per distinct argument, under the analogous scalar noise bound. Common innovations across arguments are allowed.

Suppose the complete declared learned procedure costs $F(n) + Kc_N + C(q)$ and the declared direct procedure costs $C_0(t) + Ktc_R + C(q)$, with all fitting, caching, selection and verification included. Both procedures meet the target jointly with probability at least $1 - \alpha$, and the learned procedure is strictly cheaper whenever

$$K(tc_R - c_N) > F(n) - C_0(t). \quad (9)$$

Repeated identical arguments are counted once in K . If verification or selection costs differ, their actual difference enters the right side.

The class contains nonlinear continuations represented by finite smooth neural features, not only linear economic models. The leverage bound keeps fitting

accuracy uniform as queries are reused. A bounded representation bias or trans-
 ported future adds its explicit allowance to the decision bound and reduces the
 available margin in (8). The result compares two fully specified sufficient-accuracy
 procedures; it is not a lower bound on the best possible simulation method. A Raw
 procedure may itself fit a surrogate, and then belongs to the learned comparison
 class. In particular, this corollary neither certifies the assumptions for the experi-
 ment’s jointly trained network nor asserts that its conventional surrogates must be
 slower. It provides an analytical reuse class that the numerical results need not be
 asked to establish by extrapolation.

3.4. Actual stopping and the amortization threshold

Consider a feasible scalar action in $[\ell, u]$ and a twice differentiable true objective
 with $Q'' \leq -\kappa < 0$. Define the directional residual

$$r(a) = \begin{cases} \max\{Q'(a), 0\}, & a = \ell, \\ \max\{-Q'(a), 0\}, & a = u, \\ |Q'(a)|, & \ell < a < u. \end{cases}$$

An interval enclosure of $Q'(a)$ and a lower bound on κ give a computable upper
 bound $U(a)$ in the following statement.

PROPOSITION 4—Certified stopping and complete work: *For every feasible scalar
 action,*

$$\sup_{b \in [\ell, u]} Q(b) - Q(a) \leq \frac{r(a)^2}{2\kappa}. \quad (10)$$

*For a finite sequence of candidate-fitting stages, let a service stop at the first stage whose
 mean outward bound is at most ϵ . If each bound is a deterministic enclosure for the true
 task objectives, the returned service satisfies the target regardless of the preceding failed
 checks. If checks instead have conditional error allocations α_j given the past and $\sum_j \alpha_j \leq \alpha$,
 the analogous conclusion holds with probability at least $1 - \alpha$.*

*The realized cost of a service stopping at stage J is the sum of all initialization, fitting or
 cache construction, querying, verification, and durable-record costs through J . In an exact*

fixed-plus-linear cost model $W_m(q) = F_m + qc_m$, where c_m includes marginal verification and all attempted stages, two methods that both meet the same target satisfy

$$W_N(q) < W_R(q) \iff q(c_R - c_N) > F_N - F_R. \quad (11)$$

When $c_R > c_N$, this gives the strict break-even threshold. If only cost intervals are available, a guaranteed ranking requires an upper bound for one method below a lower bound for the other, not two upper bounds.

The cost model is an interpretation, not an imposed shape for the measured frontier. Fitting stages may change with query volume, and verification can dominate marginal prediction work. We therefore execute every volume rather than extrapolate a line from a single 384-query comparison. The new experiment runs each method's frozen stage sequence; it does not search globally for a winner and then claim to have paid only that winner's bill. All comparison services and shared startup are reported separately from the service that an economist would deploy after choosing a method.

4. THE CONTROLLED ECONOMY

Let W be an m -dimensional Brownian motion on a filtered probability space satisfying the usual conditions. The state $S = (u, X)$ contains internal and external state variables. An action $a = (\theta, c, \pi, \dots)$ combines internal adjustment and external choices. The controlled state evolves as

$$dS_s = b(s, S_s, a_s) ds + \sigma(s, S_s, a_s) dW_s, \quad a_s \in \mathcal{A}(s, S_s). \quad (12)$$

The compact feasible correspondence $\mathcal{A}$ is part of the model. Actions are progressively measurable and must generate a well-defined state process satisfying the declared state restrictions. State viability does not follow from bounded actor outputs alone.

We work initially on a finite horizon T . A computational domain D is either invariant, equipped with a specified reflection mechanism, or stopped on exit. In the stopped case let $\tau = T \wedge \inf\{s \geq t : S_s \notin D\}$. The payoff $g(\tau, S_\tau)$ is specified on the entire parabolic stopping boundary, including the terminal face. For a fixed

admissible policy a , utility is the policy-specific solution

$$Y_s^a = g(\tau, S_\tau) + \int_s^\tau f(v, S_v, a_v, Y_v^a) dv - \int_s^\tau Z_v^a dW_v. \quad (13)$$

Adjustment costs are included in f . For example, time-additive utility has $f(s, S, a, y) = r_0(s, S, a) - \rho y$, where r_0 includes all current resource and adjustment costs. Define $V^a(t, x) = Y_t^{a, t, x}$ and $V^*(t, x) = \sup_a V^a(t, x)$. The value inside a candidate policy's recursive objective is its own utility process, not the unknown optimum inserted into every candidate objective.

ASSUMPTION 1—Finite-horizon comparison class: *The controlled state problem is well posed for the admissible policies used below. The coefficients are continuous, locally Lipschitz in the state uniformly over actions, and bounded on the compact stopped domain. The payoff is continuous and bounded. On an invariant utility interval containing all policy values, f is continuous in its arguments and uniformly Lipschitz in utility with constant $L \geq 0$. The admissible class is stable under concatenation and has the measurable-selection properties needed for dynamic programming. The resulting HJB problem and the fixed-policy problems obey comparison in the bounded continuous class with the specified boundary conditions.*

The comparison requirement identifies both a function class and a boundary problem. It is not asserted merely because the primitive coefficients are continuous. On a bounded, regular Dirichlet domain, a boundary barrier is one way to verify attainment of the stopping payoff. Reflection requires the corresponding Neumann or oblique derivative condition and a well-posed reflected state process. Unbounded-state applications require growth bounds and localization. These conditions are checked separately from a neural residual.

For a smooth function v , write

$$\mathcal{L}^a v = b(t, x, a)^\top Dv + \frac{1}{2} \text{tr}\{\sigma(t, x, a)\sigma(t, x, a)^\top D^2 v\}, \quad (14)$$

$$\mathcal{H}^a[v] = f(t, x, a, v) + \mathcal{L}^a v, \quad F[v] = -v_t - \sup_{a \in \mathcal{A}(t, x)} \mathcal{H}^a[v]. \quad (15)$$

The value solves $F[V^*] = 0$ with the specified boundary data, in the viscosity sense where classical derivatives do not exist. The diffusion covariance is the posi-

tive semidefinite matrix $\sigma\sigma^\top$. No claim that the product of two arbitrary positive semidefinite matrices is symmetric positive semidefinite is needed.

A continuous martingale representation can also be used when its quadratic variation is Markov in an augmented state. If $d\langle M \rangle_s = Q(s, S_s, a_s)ds$, the covariance in (14) becomes $\sigma Q\sigma^\top$. A merely predictable process Q_s is not, without further state augmentation, a deterministic coefficient of a Markov HJB equation. Quadratic variation by itself also does not imply conditionally Gaussian increments. The implementation in this paper uses the Brownian model (12).

4.1. Infinite horizon and economic normalization

For a stationary problem, the boundary-value equation is $\sup_a \{f(x, a, V) + \mathcal{L}^a V\} = 0$. We use an infinite-horizon result only after specifying admissibility, integrability, and the selected utility process. In the time-additive case, a sufficient verification requirement is that the localized Itô identities are uniformly integrable and $\mathbb{E}[e^{-\rho T} | V(S_T) |] \rightarrow 0$ for the policies being compared. Recursive problems require an appropriate utility comparison and transversality condition instead of simply borrowing the additive one.

Cardinal transformations also require care. Multiplying a fixed utility function by a positive constant does not change its maximizer. Adding a function of an *endogenously controlled preference state* generally does. Consumption units, preference normalization, and terminal utility are therefore primitive economic choices in Section A.1.3.

5. NEURAL BELLMAN OPERATORS

The actor $a_\phi(t, x)$ and critic $v_\zeta(t, x)$ represent policies and continuation values. An internal actor and an external actor can be separate parameter blocks; their joint action remains the input to the same economic generator. A smooth activation is useful for exact differentiation, while a nonsmooth neural representation can be used with the positive-weight Bellman formulation below.

5.1. Feasibility and boundary data

For a box $[a, \bar{a}]$, a sigmoid map produces interior feasible actions but does not attain its endpoints at finite weights. Binding constraints can instead be represented by a projection onto the closed interval, by an explicit boundary candidate, or by a finite action set including the endpoints. The constrained portfolio calculation uses a projection and can attain a zero risky share exactly. On a simplex, a probability map supplies a feasible interior point and boundary faces must again be considered when certifying the optimum. General constraints require a feasible parameterization, a certified projection, or a constrained action solver. A finite penalty is not an exact feasibility certificate.

On a finite horizon, an elementary hard-terminal architecture is

$$v_{\xi}(t, x) = G(x) + (T - t)N_{\xi}(t, x). \quad (16)$$

This enforces the terminal face, but not automatically a lateral boundary. A compatible lifting G_B and a factor d_B vanishing on all Dirichlet faces give $v_{\xi} = G_B + d_B N_{\xi}$. Alternatively, boundary residuals are trained separately and retained explicitly in the error bound. A reflected preference state requires its own derivative condition; a terminal architecture does not enforce it. For recursive utility, a domain-preserving transformation must additionally keep the continuation value in the domain of the aggregator.

5.2. The block-specific computational graph

Let ν be a specified distribution of state–time points and let sg denote stop-gradient. A critic step minimizes

$$L_C(\xi; \bar{\phi}) = \mathbb{E}_{\nu} \left[-\partial_t v_{\xi} - \mathcal{H}^{\text{sg}(a_{\bar{\phi}})}[v_{\xi}] \right]^2 + L_B(\xi), \quad (17)$$

where L_B is absent only for boundary conditions enforced exactly. The policy, sampling locations, and any rival actions are detached. The value, time derivative, state gradient, and Hessian remain differentiable with respect to the critic parameters.

An actor step minimizes

$$L_A(\phi; \bar{\xi}) = -\mathbb{E}_v \mathcal{H}^{a_\phi}[\text{sg}(v_{\bar{\xi}}), \text{sg}(Dv_{\bar{\xi}}), \text{sg}(D^2v_{\bar{\xi}})]. \quad (18)$$

All critic derivative channels are fixed in this step. There is no critic gradient from L_A and no actor gradient from L_C . A printed sum $L_A + \omega L_C$ may be a reporting device, but it is not jointly differentiated. The implementation exposes separate functions and regression tests for these derivative paths.

The distinction is substantive: joint differentiation can move the critic away from the policy-evaluation root. Appendix S.3 gives a counterexample, the recursive lifetime policy derivative, and the conditioning requirements. The local Hamiltonian objective is not identified with a lifetime policy gradient under arbitrary state sampling. The evaluated reference policy may remain fixed while a separate candidate actor is improved at successive work levels. Updating that reference defines a new evaluation target and is recorded explicitly.

5.3. Rollout and costate evaluation

For a frozen policy $a^{(j)}$, a positive-time Euler rollout from (t, y) supplies a sample return $\widehat{V}^{(j)}(t, y)$ and its pathwise derivative $\widehat{p}^{(j)}(t, y)$. Differentiation includes the state dependence of subsequent policy actions. Freezing the policy means freezing its parameters, not deleting that derivative. The targets evaluate the current policy, not the optimal policy, and contain no maximizing action labels. For fixed finite weights, bounded actions and smooth bounded production justify pathwise differentiation of this finite rollout away from the harmless clipping kinks. These are derivatives of the discretized evaluation problem, not asserted unbiased derivatives of its continuous-time limit.

The critic minimizes

$$\widehat{\mathbb{E}} \left[0.05 \{v_\theta - \widehat{V}^{(j)}\}^2 + \|d(D_y v_\theta - \widehat{p}^{(j)})\|_d^2 \right]. \quad (19)$$

The factor d puts normalized marginal values on an order-one scale. The rollout implementation uses two hidden tanh layers and a hard terminal component.

Holding its new derivative fixed, the actor then maximizes

$$\hat{\mathbb{E}} \left[\overline{\log a_\phi} - \frac{\zeta}{2} \bar{a}_\phi^2 - a_\phi \cdot \text{sg}(D_y v_\theta) \right]. \quad (20)$$

Terms independent of actions have been omitted. Consumption is represented directly by $m_0(t) + \epsilon \tanh N_\phi(t, y)$; there is no online Hamiltonian search in its deployment. Value evaluation, costate evaluation, and feasible improvement are distinct operations, with their own stored losses and work counts. The number of critic and actor steps per rollout batch is a declared design parameter. Failure of either numerical block is recorded, not interpreted as successful policy improvement.

The stochastic gradient targets make the economically relevant marginal continuation values visible to the evaluation loss. This is not a new Bellman principle, nor an assumption that derivative regression invariably dominates differential residual fitting. It is a concrete implementation of the NBO evaluation map. The preserved differential implementation and its two independent trace banks continue to test a different evaluation procedure. Both feed the same economic distinction between policy evaluation and feasible improvement.

5.4. The implemented map and its verification output

Fix the economy, boundary realization, network classes, optimization budget, and verification procedure, denoted collectively by ϑ . Let $\mathcal{P}$ be the admissible feedback policies, Ξ the critic parameter space, and Ω the sampling streams. The implemented NBO is the numerical map

$$\mathcal{N}_\vartheta : \mathcal{P} \times \Xi \times \Omega \longrightarrow \mathcal{V}_\vartheta \times \mathcal{P}_\vartheta \times \mathcal{C}_\vartheta, \quad (a, \xi, \omega) \longmapsto (v, \tilde{a}, C). \quad (21)$$

The first component is approximate policy evaluation; the second is feasible improvement; the third records verification of the resulting pair. A record identifies the model, parameters, domain, boundary faces, evaluation error, maximization error, coverage, discretization errors, and unchecked assumptions. An unavailable bound is not zero. Failure to obtain a useful certificate is a possible output of the map. With its sampling stream fixed, the implementation is deterministic. We do

not assert that a network learns an unrestricted Bellman operator on a function space.

The evaluation–improvement structure is an instance of approximate policy iteration, including its partial-evaluation variants (Scherrer et al., 2015). Neural representation does not introduce a new principle of optimality, and monotonicity belongs to the underlying positive-weight scheme, not to gradient descent. The NBO specification joins the block-specific derivative graph to boundary and utility-domain restrictions and to a common economic error account. Sections S.11.1 and A.1.3.4 implement the differential and positive-weight versions of this map, respectively, with independent verification of their outputs.

5.5. *Evaluation, feasible improvement, and approximation*

The algorithm produces a continuation approximation and a feasible policy. Its economic interpretation requires an error account for each. The general results in technical Section S.1 bound the consequence of a fixed-policy evaluation error and an incomplete feasible maximization under the stated comparison and boundary conditions. The positive-weight Bellman formulation also provides a monotone approximation route when classical derivatives are unavailable. The complete statements and proofs remain current in the supplement.

The following capital economy permits a more operational account. Its payoff identity follows the implemented action through the original diffusion. Its finite Bellman identity measures continuation error along the candidate’s actual action change. This specialization makes it possible to compare a learned continuation with a cached simulation continuation using the same economic decision and a common assessment design.

5.6. *A common continuation for different economic decisions*

The capital economy permits a direct study of continuation reuse. Consider a finite-period decision in which a temporary consumption weight, consumption externality, or consumption cap changes at the current date. Subsequent preferences, technology, Gaussian innovations, terminal dispersion cost, and reference

Algorithm 1 Neural Bellman evaluation, feasible improvement, and economic assessment

Require: Economic primitives and information; feasible actions and utility domain; boundary data; evaluation variant; training, validation and stopping rules; independent final assessment.

- 1: Declare the feasible reference policy to be evaluated. Initialize a feasible candidate and a boundary-compatible continuation representation.
 - 2: **for** each permitted evaluation-improvement iteration **do**
 - 3: Freeze the declared reference policy, rivals and evaluation target. Fit only the critic by the declared differential, positive-weight, or rollout rule. The reference may remain fixed across work levels.
 - 4: Freeze the critic and its value/derivative channels. Update only the feasible actor, or apply the declared feasible action solver.
 - 5: Record all evaluation and improvement work, corrections, unsuccessful checks, and the resulting deployment rule.
 - 6: Apply the predetermined validation and stopping rule, using no final-test outcomes.
 - 7: **end for**
 - 8: Freeze the selected policy and continuation object before independent final evaluation.
 - 9: Compute the specified economic endpoints and direct contrasts, with their approximation, implementation and statistical accounts.
 - 10: Return the fitted objects, their implementation and domain, and the complete accuracy and work record.
-

consumption remain fixed. For task $\theta = (w, \eta, u)$, the action-dependent payoff is

$$Q_\theta(y, a) = A_0\{\overline{w \log a} - \eta \bar{a}^2/2\} - W_0 \bar{a} + S\{x_a(y)\},$$

$$x_a(y) = y + h\{c1 + f(y) - a\}, \quad S(x) = \mathbb{E}F(x; \xi), \quad a \in [\underline{a}, u]^d. \quad (22)$$

Here a known current term independent of a is suppressed. The random variable $F(x; \xi)$ includes the first Gaussian innovation at postdecision mean x , all subsequent

reference rewards, and the terminal payoff. The stored period weights and innovation amplitudes define this finite capital problem. Its decision values have their own interpretation, separate from the continuous-diffusion value in the preceding payoff theorem.

Proposition S.14 establishes that S is identical across these tasks. NBO therefore fits one scalar continuation and queries its gradient when the current economic problem changes. This reuse has a precise information boundary. A stored payoff or costate remains an exact rollout observation at the argument where it was computed. Reusing the same innovation array at another argument requires reevaluating its state recursion and payoff. The Raw comparators may cache valid observations and innovations under the same rule. The comparison concerns the accuracy and complete cost of these representations of the common continuation.

The scalar representation also imposes structure on costate fitting. Fix postdecision arguments and positive weights before drawing rollout labels. Stack the true gradients, multiplied by the square roots of the weights, in q . Let the corresponding labels be $z = q + \varepsilon$, with $\mathbb{E}\varepsilon = 0$ and finite covariance Σ . For differentiable scalar features fixed independently of these labels, let P be the Euclidean orthogonal projection onto their stacked weighted gradient columns. A least-squares scalar critic has fitted gradients Pz , and

$$\mathbb{E}\|Pz - q\|^2 - \mathbb{E}\|z - q\|^2 = \|(I - P)q\|^2 - \text{tr}\{(I - P)\Sigma\}. \quad (23)$$

Proposition S.15 proves the identity without requiring independent coordinates or Gaussian label noise. The variance removed outside the gradient dictionary is the statistical benefit; approximation error is its cost. When the true gradient belongs to the dictionary and the removed variance is positive, the scalar critic strictly reduces this risk. The same fitted object can then serve every task in (22).

This calculation gives a conditional statistical account of the scalar critic. A fully trained nonlinear network generally chooses its features using its labels and need not coincide with this fixed projection. A Raw procedure using the same projection has the same identity. The economic comparisons consequently evaluate all five complete procedures on independent payoff observations, including their

approximation errors, action choices, and computational work. Technical Section S.8 gives the full proofs, direct conditional payoff range, and separate accuracy account over an entire scalar intervention interval.

5.7. Signed continuation assessment

The occupation account can retain the sign of the continuation prediction error. At a sampled candidate state and date, evaluate the reference continuation from the candidate and reference postdecision states using the same future innovations. Add the current reward difference before forming the interval. The conditional expectation is the reference Bellman advantage; averaging over the candidate's occupation distribution gives the finite policy gain.

Proposition S.12 makes this statement precise. The predicted improvement and its signed continuation error cancel sample by sample. A lower endpoint for their combined statistic therefore need not pay separately for the full range of the fitted predictor. Proposition S.13 supplies an economic range using the action tube, bounded production derivatives, and antithetic future paths. It retains the original clipping and arithmetic account and the separate transfer to the continuous-time payoff. The complete derivation is in technical Section S.7.7.

This calculation addresses the precision of an economic identity for the selected policy. The comparison of a learned critic with a Raw continuation remains a distinct empirical question. The repeated-decision design makes that comparison directly, including the cost and benefit of reusing each continuation representation.

6. COMPUTATIONAL DESIGN

The experiments distinguish a decision problem, a complete algorithm, a selected policy, and an independent payoff bank. Each confidence statement conditions on the objects fixed before its bank. The earlier complete method distribution contains sixteen equally weighted algorithm streams at each dimension. Generating new payoff paths for those policies does not create new training configurations. A finite repeated-decision economy likewise has its own value target and is not silently identified with the continuous diffusion.

6.1. *Reusable continuation and new decisions*

The central comparison studies the common continuation in (22) across twelve temporary preference, externality, and consumption-cap tasks. Each task is evaluated at thirty-two fixed states, giving 384 equally weighted queries in each dimension. Four capital calibrations retain the original low-volatility primitives, a two-year quarterly horizon, a four-year half-year horizon, and an intermediate three-year quarterly horizon. The two- and four-year calibrations each have eight decision periods. The new model and its stored period coefficients define a finite repeated-decision economy; no continuous-time transfer is asserted for these menu values.

Five complete procedures share the permitted continuation information: the NBO scalar critic, an unrestricted vector costate predictor, a materialized Raw actor, a direct-policy actor, and query-level Raw sample-average optimization. The vector predictor uses the same postdecision labels as NBO. The Raw actor retains labels at reference actions, whereas direct policy and query-level Raw may retain innovations and reevaluate the continuation when their actions change. Thus the comparisons assess complete representation and optimization procedures, with all approximation and action-selection errors included.

Each calibration and dimension has sixteen streams fixed before confirmation. All methods retain three work stages, using 128, 512, and 2,048 replay states for the fitted procedures and 4, 16, and 64 antithetic paths for query-level Raw. No stage is selected on the final payoff bank. The 216 primary statements cover every method's reference gain and every direct NBO contrast at all stages and cells. Independent confirmation uses 512 antithetic pairs per query and family error 0.018. The fixed materiality and equivalence margins are both 10^{-4} ; a positive interval can also lie inside the equivalence band. A separate family with error 0.002 assesses 128 scalar-intervention accuracy bounds. Technical Section S.19 gives the complete calibration, fitting, inference, and work contracts.

6.2. Prospective confirmation of previously selected policies

The confirmation retains all 128 frozen policies from the previous experiment: four methods, sixteen declared streams, and two dimensions. Before generating its new bank, it fixes the original paired-transfer theorem, outward constants, clipping account, reporting code, and an independent family error allocation of 0.01. The new bank has 8,192 paths on 2,048 decision cells. It is independent of the original assessment bank and of the completed training. Its target is the mean payoff of the original finite policy distribution and its specified direct comparisons.

The deterministic transfer is audited separately in technical Section [S.7.6](#). The confirmation retains the original protected constants. It does not substitute a smaller high-precision diagnostic for an interval enclosure. Earlier online stopping decisions remain those made on their original checkpoint banks, irrespective of later confirmation.

6.3. A strengthened neural-HJB implementation

This separate continuous-economy family trains new policies at four capital calibrations—high volatility, the original low volatility, a longer horizon, and joint economic stress—in dimensions ten and fifty. Each cell has eight new streams, giving 64 trials, 256 method-group processes, and 320 policy outputs. A new 8,192-path, 2,048-cell bank assesses each trial. All 648 conditional stream and finite-stream-mean events share error allocation 0.01. The stress design changes coupling, terminal dispersion preferences, initial heterogeneity, and the feasible actor radius together; its comparison does not isolate the effect of any one primitive. This family is distinct from both the fixed-policy replication and the sixteen-stream finite menu.

The stronger comparator evaluates the diffusion trace exactly at dimensions ten and fifty. Four learning-rate and collocation configurations are fixed in advance. Each uses Adam followed by L-BFGS; independent residual diagnostics select the configuration and its checkpoint. The final payoff bank does not select the fitted network. The implemented policy records both an optimized action root and a distilled actor, with separately reported payoff and work for each output. Complete work includes every fitted configuration, selection, distillation, action

deployment, and verification. This is a fixed-envelope comparison: a stop based on independent residual checks is not an online economic-target stopping rule. The previous stochastic-trace implementation and its complete outcomes remain in technical Section S.14.

6.4. Precision of the signed occupation account

The signed Bellman experiment uses independent candidate-occupation and continuation banks. Its protected statistic combines the current reward and paired future-return difference before clipping and inference. Its allocation is fixed using the proved economic range and a power calculation. It reports the numerical transfer and every remaining allowance alongside the stochastic radius. The result is interpreted as an assessment of the selected policy’s Bellman account; direct critic–Raw comparisons retain their own payoff, continuation-risk, and work estimands.

6.5. Economic and inferential scope

The confidence family is stated separately for each new experiment. A union of results from different families is asserted only with its explicit allocation. Previous publication evidence retains its original probability account. All results retain their economic parameters, state population, implemented controller, and complete cost boundary. A direct comparison interval that spans the declared margin remains unresolved. A protected lower gain over the schedule, a bound on full-class regret, a finite-grid residual, and an empirical refinement discrepancy answer different economic questions.

7. ECONOMIC VALUE AND COMPUTATIONAL WORK

7.1. The value of a reusable continuation

The complete menu experiment gives a positive economic role for the learned continuation at a stated computational budget. All 640 method fits and 1,920 stage candidates are retained; no fit failed and no stage used the fallback. Table II reports every final-stage direct comparison, pooling the sixteen declared streams and all 384 task-state queries within each of the eight cells. Its units are 10^{-4} , the fixed

materiality and equivalence margin. The intervals belong to the same 216-event family as the complete earlier-stage record.

TABLE II
INCREMENTAL VALUE OF THE REUSABLE NBO CONTINUATION

Economy	d	NBO – Vector	NBO – Raw actor	NBO – DPO	NBO – SAA
Original	10	[0.00, 0.01]	[0.06, 0.07]	[0.06, 0.07]	[-0.01, 0.01]
Original	50	[0.03, 0.05]	[0.25, 0.26]	[0.24, 0.25]	[-0.03, -0.02]
Quarterly	10	[-0.02, 0.02]	[0.20, 0.26]	[0.12, 0.17]	[-0.06, -0.01]
Quarterly	50	[0.22, 0.27]	[0.82, 0.90]	[0.76, 0.83]	[-0.41, -0.36]
Long	10	[0.14, 0.92]	[5.67, 6.70]	[-1.03, -0.19]	[-1.99, -1.22]
Long	50	[16.38, 18.13]	[18.02, 19.72]	[3.27, 4.25]	[-2.68, -1.89]
Intermediate	10	[-0.01, 0.03]	[0.25, 0.33]	[0.15, 0.21]	[-0.03, 0.02]
Intermediate	50	[0.66, 0.79]	[1.47, 1.63]	[1.30, 1.44]	[-0.49, -0.39]

Notes: Intervals are in units of 10^{-4} and use the final, third work stage. Every cell averages all sixteen declared streams and all 384 fixed current-period queries. Each query has 512 independent antithetic future pairs; opposite signs form one observation. Five methods share each pair. All three stages and all eight economic-dimension cells contribute to one 216-event family with error probability 0.018. Current reward and a known terminal quadratic are removed before the random residual is bounded. The point estimator underlying each interval is the clipped residual estimator plus this known offset. Numerical arithmetic, clipping and the unclipped Gaussian law are included. The materiality margin is 10^{-4} .

In both previously unpiloted fifty-dimensional designs, NBO yields materially higher mean payoff than the materialized Raw and direct-policy actors. Its lower gain over the Raw actor is 0.001802 in Long and 0.000147 in Intermediate; the corresponding lower gains over direct policy are 0.000327 and 0.000130, all rounded downward. Each exceeds 10^{-4} . In Long at dimension fifty, NBO also exceeds the fitted vector costate by at least 0.001638. These comparisons measure the economic consequences of using the complete scalar continuation procedure. They do not estimate an independent critic-risk difference or isolate variance removal from nonlinear approximation and optimization.

The payoff improvements over direct policy in those two fifty-dimensional cells accompany lower measured construction and query work. NBO's mean complete three-stage clocks are 14.59 seconds against 38.03 for direct policy in Long, and

TABLE III
COMPLETE CONSTRUCTION AND QUERY WORK

Economy	d	NBO	Vector	Raw actor	DPO	Cached SAA
Original	10	12.52	8.73	6.86	16.21	15.60
Original	50	12.67	9.85	7.07	20.24	27.68
Quarterly	10	10.36	7.69	6.09	21.85	25.23
Quarterly	50	14.34	11.15	8.28	38.93	62.90
Long	10	12.05	8.87	6.96	26.53	30.42
Long	50	14.59	11.06	8.16	38.03	67.04
Intermediate	10	11.96	9.16	7.01	36.55	44.71
Intermediate	50	12.70	9.62	7.18	45.58	79.46

Notes: Seconds per stream, averaged over all sixteen streams, including unsuccessful fits. Each entry is the actual outer-process clock for all three stages, imports, source checks, reusable training data, optimizer and derivative work, all saved queries and durable output. Fixed dependency installation and source extraction precede the clock. Independent confirmation is a shared measured bundle, reported separately in the complete account; the table does not allocate one-fifth of that bundle as an observed standalone verification cost.

12.70 against 45.58 in Intermediate. The materialized Raw actor and vector predictor have lower construction and query clocks than NBO in every final cell. Thus the economic increment over these two representations has a computational cost. The comparison with direct policy instead improves both the protected payoff and the observed work at the stated final budget in the two specified cells.

Cached sample-average optimization supplies a further comparison with direct simulation. In the six final cells outside Long, the complete NBO–SAA interval lies within $[-10^{-4}, 10^{-4}]$. NBO uses less measured construction and query work than the final sixty-four-path SAA in every one of these cells. The reductions range from 19.73 to 84.02 percent of the SAA mean clock. This is a comparison of the prespecified final-stage procedures. It does not identify the minimum work required for a given accuracy: the complete record also retains the cheaper four- and sixteen-path SAA stages.

The Long calibration displays the corresponding accuracy tradeoff. SAA has a protected advantage exceeding 10^{-4} over NBO in both dimensions, at higher final construction and query work. In Long at dimension ten, direct policy also has a

positive protected advantage over NBO, although its material superiority at 10^{-4} is unresolved. Across all three stages, NBO has nine material lower endpoints against Vector, seven against the Raw actor, four against direct policy, and none against SAA, out of twenty-four comparisons with each. The respective numbers of strictly negative NBO upper endpoints are zero, three, seven, and nineteen. The complete intervals in technical Section S.19.5 retain every reversal and every unresolved result. Positive differences and practical equivalence are reported as overlapping conclusions when both inequalities hold.

Independent confirmation is a shared computation. Its complete mean bill ranges from 10.73 to 82.20 seconds per stream across the eight cells; Table S.34 gives every cell and the full clock range. Adding that same bill to each method preserves the final-cell work ordering, but the resulting total is an explicit conservative allocation of the complete five-method, three-stage audit. It is not the time of an independently run single-method verifier. The timing comparisons describe the measured finite collection of runs and carry no unobserved-runtime confidence claim.

TABLE IV

CONDITIONAL ACCURACY ON THE PRESCRIBED SCALAR INTERVENTION

Economy	d	Gap $\leq 10^{-4}$	Smallest gap bound	Largest gap bound
Original	10	16/16	0.0000002	0.0000002
Original	50	16/16	0.0000002	0.0000004
Quarterly	10	16/16	0.0000016	0.0000024
Quarterly	50	16/16	0.0000138	0.0000141
Long	10	0/16	0.0440599	0.0454588
Long	50	0/16	0.1962975	0.1963168
Intermediate	10	16/16	0.0000023	0.0000024
Intermediate	50	0/16	0.0002819	0.0002846

Notes: All 128 conditional statements are retained under a separate error probability of 0.002, with 65,536 independent antithetic pairs for each fixed scalar candidate. Each upper bound covers the whole continuous scalar action segment at the predeclared fixed high-dimensional state. It is not a bound on the complete vector action problem or continuous-time adapted policies. Endpoints are rounded upward.

The scalar exercise provides a separate absolute-accuracy result. Eighty of the 128 fixed candidates have a protected gap below 10^{-4} over their entire continuous scalar intervention interval. All sixteen streams attain this criterion in Original and Quarterly at both dimensions and in Intermediate at dimension ten. Long at both dimensions and Intermediate at dimension fifty retain larger gap upper bounds. Nonattainment of an upper certificate does not establish a corresponding lower bound on actual suboptimality. These statements concern the fixed current scalar intervention with the common future reference policy; they do not assert near-optimality over all vector actions or adapted continuous-time controls.

7.2. Independent payoff confirmation

The independent confirmation evaluates all 128 previously selected policies, without retraining or further selection. Table V reports the schedule-relative method gains and the associated full-adapted-class regret bounds. Table VI reports every direct NBO comparison under both the original and the prospectively fixed paired transfer. All eight method endpoints and six direct contrasts belong to the declared 238-event family, with total error allocation 0.01.

The lower endpoints for NBO minus the original neural-HJB implementation are 0.000176 in dimension ten and 0.001342 in dimension fifty, rounded downward. Both exceed the prespecified materiality margin 10^{-4} . In dimension ten the original transfer still gives an interval containing zero on these new observations. Thus the new experiment confirms the value of the sharper paired account under a proof, constants, and reporting rule fixed before its bank. The earlier use of that account remains a post-freeze analytical reanalysis of the earlier observations.

The four NBO–Raw and NBO–direct-policy intervals remain unresolved at 10^{-4} . Their point estimates are close, but the protected intervals do not establish practical equivalence. This confirmation therefore supports the stated comparison with the original HJB implementation and the positive schedule-relative gains; the reusable-continuation experiment supplies the separate comparison of the critic’s incremental economic value. The full-class regret bounds in Table V retain the anchor gap.

TABLE V

INDEPENDENT R16 CONFIRMATION OF THE FIXED R15 POLICIES

d	Method	Mean gain	Simultaneous interval	Regret bound
10	NBO	0.001269	[0.000943, 0.001596]	0.025971
10	Raw	0.001268	[0.000942, 0.001595]	0.025972
10	Direct policy	0.001264	[0.000937, 0.001590]	0.025977
10	Neural HJB	0.000974	[0.000646, 0.001302]	0.026268
50	NBO	0.001342	[0.000955, 0.001729]	0.036845
50	Raw	0.001343	[0.000956, 0.001730]	0.036843
50	Direct policy	0.001347	[0.000960, 0.001734]	0.036840
50	Neural HJB	-0.000155	[-0.000560, 0.000249]	0.038359

All 128 original R15 selected policies are evaluated without retraining on new R16 banks. Each of the sixteen original streams receives 8192 independent paths on 2048 cells. The paired theorem, constants and new familywise allocation $\alpha = 0.01$ precede these observations. Interval endpoints and regret upper bounds are rounded outward. This is independent confirmation of fixed policies; historical stopping is unchanged.

TABLE VI

PROSPECTIVELY FIXED PAIRED TRANSFER ON FRESH CONFIRMATION BANKS

d	NBO minus	Mean	Original transfer	Fixed paired transfer
10	Raw	0.000001	[-0.000365, 0.000367]	[-0.000114, 0.000115]
10	Direct policy	0.000005	[-0.000361, 0.000372]	[-0.000109, 0.000120]
10	Neural HJB	0.000295	[-0.000076, 0.000667]	[0.000176, 0.000415]
50	Raw	-0.000001	[-0.000432, 0.000429]	[-0.000136, 0.000133]
50	Direct policy	-0.000005	[-0.000436, 0.000426]	[-0.000140, 0.000130]
50	Neural HJB	0.001497	[0.001046, 0.001948]	[0.001342, 0.001653]

All 128 original R15 selected policies are evaluated without retraining on new R16 banks. Each of the sixteen original streams receives 8192 independent paths on 2048 cells. The paired theorem, constants and new familywise allocation $\alpha = 0.01$ precede these observations. Interval endpoints and regret upper bounds are rounded outward. This is independent confirmation of fixed policies; historical stopping is unchanged.

The new per-policy certificates attain the primary schedule-gain target 0.0005 for 14 of 16 NBO policies in dimension ten and all 16 in dimension fifty. The corresponding counts are 13 and 16 for Raw, 10 and 16 for direct policy, and zero in both dimensions for the original HJB implementation. These are outcomes of a new final assessment. They leave the original online stopping decisions unchanged. The pooled dimension-ten HJB lower endpoint exceeds 0.0005, although no individual HJB policy has a new protected lower endpoint exceeding that target. Technical Section S.20 gives the complete target account and additional verification work.

7.3. Strengthened neural-HJB comparison

The strengthened comparison trains each procedure afresh in four continuous diffusion economies, at dimensions ten and fifty, with eight fixed streams per cell. All 64 trials, 256 construction processes, and 320 policy outputs are retained. The exact-trace HJB procedure fits all four declared configurations, chooses its network using independent residual observations, and deploys both the optimized greedy action and its distilled actor. No training failure or reference-policy fallback occurred.

All forty method–economy mean-gain lower endpoints exceed the primary schedule-relative target 0.0005. Table S.43 in the technical supplement gives all gains beside their full-adapted-class regret bounds. These positive policy improvements extend across the declared changes in diffusion risk, horizon, and joint stress. The remaining anchor gap is appreciable, particularly in the longer-horizon and joint-stress economies, so these intervals do not establish 10^{-4} accuracy relative to the adapted optimum.

Table VII reports all 32 direct NBO comparisons. In the low-volatility, fifty-dimensional economy, the lower endpoints against HJB’s greedy and distilled policies are 0.000334 and 0.000340, rounded downward. Both exceed 10^{-4} . The other thirty intervals contain zero and do not establish practical equivalence at that margin. In particular, the comparisons against the strengthened HJB implementation in the original high-volatility economy are unresolved in both dimensions.

This result is distinct from the prospective replication against the original HJB implementation in the preceding subsection.

TABLE VII: Direct common-path NBO comparisons across economic designs

Design	d	NBO minus	Mean	Fixed paired interval
high	10	Raw	-0.000000	[-0.000180, 0.000180]
high	10	Direct policy	0.000004	[-0.000176, 0.000185]
high	10	HJB greedy	0.000008	[-0.000173, 0.000189]
high	10	HJB actor	0.000010	[-0.000171, 0.000191]
high	50	Raw	-0.000002	[-0.000202, 0.000197]
high	50	Direct policy	-0.000001	[-0.000201, 0.000198]
high	50	HJB greedy	-0.000020	[-0.000221, 0.000180]
high	50	HJB actor	0.000040	[-0.000161, 0.000240]
low	10	Raw	0.000000	[-0.000153, 0.000153]
low	10	Direct policy	0.000002	[-0.000151, 0.000155]
low	10	HJB greedy	0.000024	[-0.000129, 0.000177]
low	10	HJB actor	0.000024	[-0.000130, 0.000177]
low	50	Raw	-0.000000	[-0.000174, 0.000174]
low	50	Direct policy	0.000002	[-0.000172, 0.000176]
low	50	HJB greedy	0.000515	[0.000334, 0.000697]
low	50	HJB actor	0.000522	[0.000340, 0.000704]
long	10	Raw	0.000002	[-0.000901, 0.000905]
long	10	Direct policy	0.000020	[-0.000883, 0.000924]
long	10	HJB greedy	0.000060	[-0.000846, 0.000965]
long	10	HJB actor	0.000072	[-0.000834, 0.000977]
long	50	Raw	-0.000003	[-0.001159, 0.001153]
long	50	Direct policy	0.000006	[-0.001151, 0.001162]
long	50	HJB greedy	0.000263	[-0.000897, 0.001423]
long	50	HJB actor	0.000323	[-0.000836, 0.001483]
stress	10	Raw	0.000001	[-0.001038, 0.001040]
stress	10	Direct policy	0.000007	[-0.001033, 0.001047]
stress	10	HJB greedy	0.000021	[-0.001020, 0.001062]

Design	d	NBO minus	Mean	Fixed paired interval
stress	10	HJB actor	0.000038	[-0.001003, 0.001079]
stress	50	Raw	-0.000003	[-0.001254, 0.001247]
stress	50	Direct policy	0.000019	[-0.001233, 0.001271]
stress	50	HJB greedy	-0.000024	[-0.001277, 0.001229]
stress	50	HJB actor	0.000202	[-0.001051, 0.001455]

The continuous-time paired transfer and all calibration-specific constants are fixed before confirmation. Original-transfer intervals remain in the complete machine-readable record. A crossing interval is unresolved, not evidence of equivalence.

The complete fixed-envelope costs also change with the implementation. In the low-volatility, fifty-dimensional cell, mean complete work is 102.27 seconds for NBO, 297.44 for HJB's greedy deployment, and 240.86 for its distilled actor. Raw takes 91.29 seconds there, and its economic comparison with NBO remains unresolved. Table S.44 retains all forty work rows and every final target outcome. Each HJB deployment is charged the common four-candidate construction, selection, distillation, and finalization, together with its own deployment. These are measured fixed-envelope costs; no new online work-to-target stopping experiment is inferred from them.

Technical Section S.21 reports the full economic grid, all four configurations, independent residual diagnostics, trace variance checks, and action diagnostics. Exact derivatives eliminate stochastic trace noise from training the specified network. The held-out diagnostics assess finite collocation banks, and successive value and policy changes compare iterates. Neither quantity is a global value-error certificate. The family contains no direct greedy-minus-distilled payoff event, so the two deployments' separate payoff and work records are not interpreted as a protected distillation-loss comparison. The result concerns these declared procedures and the complete eight-stream populations in each economy.

7.4. A prospective signed occupation account

The new assessment evaluates every original selected NBO policy on fresh occupation paths. Its proof, clipping interval, confidence family, and 1,024 bridges per stream were fixed before the new bank was drawn. The paired continuation calculation uses the same innovations at the candidate and reference action endpoints. It evaluates the signed continuation correction and applies Proposition S.12 directly to their joint Bellman gain, rather than subtracting separately estimated endpoints.

TABLE VIII

A FRESH SIGNED BELLMAN MECHANISM ACCOUNT

d	M mean	C mean	L_G	Hold/transfer	Economy lower
10	0.001243	-0.000002	0.001067	0.000300	0.000767
50	0.001354	0.000001	0.001138	0.000363	0.000776

Notes: Each dimension fully enumerates the original sixteen selected NBO streams, with 1,024 independent occupation bridges per stream. One bank of two antithetic pairs supplies the common-noise continuation endpoints. The fitted scalar continuation cancels in $G = M - C$ before inference. All six two-sided (d, M, C, G) statements share a new familywise error probability of 0.01. The finite lower endpoint includes numerical sample and prefix-representation allowances. The final column additionally retains the original holding deficit and full continuous-payoff transfer. This result does not rank the critic against Raw.

In dimension 10, the signed occupation account certifies a continuous-economy gain of at least 0.000767.

In dimension 50, the signed occupation account certifies a continuous-economy gain of at least 0.000776.

The original negative Cauchy–Schwarz mechanism bounds remain part of the record. The present calculation changes the sufficient error account and uses an independently frozen confirmation bank. Its positive endpoints, when obtained, concern the occupation Bellman identity of the returned policy. They do not establish lower prediction risk than Raw or identify a unique incremental contribution of the learned critic. The separate common-data method comparison addresses that question. Both signed lower endpoints exceed the primary target 0.0005; neither attains the secondary target 0.001. The intervals for the signed continuation

correction itself still contain zero. The positive economic conclusion uses the joint Bellman-gain statistic, whose paired future innovations preserve the cancellation in Proposition S.12. The menu comparison measures economic decision values and work. Section 8 supplies the separate independent assessment of continuation-risk differences; the payoff comparisons alone do not imply it.

8. CONTINUATION RISK AT THE ECONOMIC ACTION

The payoff comparison establishes the economic performance of the complete NBO procedure. To assess the fitted continuation itself, we compare its prediction with a cached simulation predictor at the same action pair. For query j , let x_j^+ be the postdecision state of the final NBO action and x_j^0 that of the reference action. The object to be predicted is $c_j = S(x_j^+) - S(x_j^0)$ in the finite economy of Section 5.6. The scalar network supplies $p_{Nj} = \hat{S}_N(x_j^+) - \hat{S}_N(x_j^0)$; cached Raw supplies p_{Rj} . Both are evaluated at these same states. Conditional on the complete training record, these states, and the realized prediction caches, define

$$R_M = J^{-1} \sum_{j=1}^J (p_{Mj} - c_j)^2, \quad M \in \{N, R\}. \quad (24)$$

Conditioning permits nonlinear fitting and learned features. It does not replace the fitted network by a fixed linear dictionary.

A fresh common-future-path contrast Z_{jk} has conditional mean c_j . The difference of its two squared prediction errors is

$$T_{jk} = (p_{Nj} - p_{Rj})(p_{Nj} + p_{Rj} - 2Z_{jk}). \quad (25)$$

Its conditional mean averages to $R_N - R_R$, and its conditional variance is $4(p_{Nj} - p_{Rj})^2 \text{Var}(Z_{jk})$. The common squared-noise term cancels exactly. Centering the remaining continuation residual by its known terminal quadratic component gives a bounded statistic with explicit clipping and arithmetic allowances. Propositions S.26 and S.28 give the complete identities and finite-sample interval. The same scalar differences define a population contrast-training loss (Corollary S.27); the policies reported here are the unchanged fitted policies, not policies retrained with that loss.

The independent assessment fixes all predictions and geometric bounds before drawing its audit innovations. It retains four calibrations, both dimensions, all sixteen streams, and all 384 queries per stream. Each query receives 256 independent antithetic pairs. A sign pair is averaged into one observation; the same audit is reused across the three Raw budgets and is not counted three times as independent information. There are 12,582,912 distinct pair observations over the eight cells. The comparison conditions on each realized Raw cache, rather than averaging over unexecuted caches. It also conditions on NBO's chosen action pairs; it is not an occupation-risk claim for every feasible action or for Raw's separately chosen actor.

TABLE IX
CONDITIONAL CONTINUATION-RISK DIFFERENCES

Design	d	Raw: 4 paths	Raw: 16 paths	Raw: 64 paths
Low	10	$[-0.0242, 0.00950]$	$[-0.00938, 0.00650]$	$[-0.00491, 0.00486]$
Low	50	$[0.00817, 0.0552]$	$[0.0106, 0.0564]$	$[0.0114, 0.0566]$
Quarterly	10	$[-104, -62.7]$	$[-25.8, -4.52]$	$[-5.94, 8.78]$
Quarterly	50	$[-38.0, -14.9]$	$[-10.2, 3.59]$	$[-2.68, 7.75]$
Long	10	$[-8360, -4760]$	$[-1650, 619]$	$[125, 1900]$
Long	50	$[-5220, -2690]$	$[-1300, 27.5]$	$[-326, 654]$
Intermediate	10	$[-151, -73.0]$	$[-43.7, -4.20]$	$[-12.9, 10.4]$
Intermediate	50	$[-72.2, -15.1]$	$[-20.5, 13.5]$	$[-4.37, 17.2]$

Notes: All intervals are for $R_N - R_R$, in units of 10^{-8} . Negative intervals favor NBO. Endpoints are rounded outward to three significant digits; the complete precision is retained in the report. The 24-event family has simultaneous coverage at least .99. Predictions, action pairs, and realized Raw caches are fixed before the independent assessment.

Table IX reports all 24 comparisons, not only those favoring learning. Eight intervals have upper endpoint below -10^{-8} and establish lower NBO contrast risk. Four have positive lower endpoint and favor Raw; twelve include zero. At the four-path budget, every Quarterly, Long, and Intermediate cell favors NBO. At sixteen paths, the ten-dimensional Quarterly and Intermediate cells also favor NBO. In contrast, the fifty-dimensional Low cell favors Raw at every budget, and Long in ten dimensions favors the 64-path Raw predictor. This pattern identifies where the

observed fitted continuation improves prediction, and where additional simulation removes that advantage.

The preassigned risk margin 10^{-8} is the square of the payoff unit 10^{-4} , not a payoff difference. Proposition S.25 bounds average binary-decision loss by $\sqrt{R_M}$, but a smaller risk need not give a better action: prediction errors with different signs can reverse that ordering. A difference of risks is not an absolute-risk certificate. Accordingly, the risk result and the actual five-method payoff result provide distinct evidence for NBO; neither is substituted for the other.

The risk family has failure probability at most .01. Combining it with the menu payoff family, whose allocation is .018, gives simultaneous coverage at least .972 by a union bound, without requiring independence between families. Combining it instead with all four earlier scientific families, including their scalar events, gives coverage at least .94. The individual family guarantees remain unchanged. The source freeze, pre-audit receipts, and exact report replay appear in Section S.24.

9. DECISION ACCURACY ACROSS THE COMPLETE BUDGET CATALOGUE

A final-budget comparison alone does not determine which computed procedure meets an economic tolerance at the lowest cost. We therefore use every prespecified menu stage. In each economic cell, the catalogue contains the reference and the five methods at each of three stages. Write V_i for candidate i 's mean payoff over the same complete sixteen-stream population and 384 queries. The target in this section is $g_i = \max_j V_j - V_i$ over these sixteen candidates. This is a catalogue of computed vector decisions, not the full continuous vector-action set or the adapted diffusion class.

The original simultaneous intervals bound five reference-relative gains and four direct NBO contrasts at each stage. If $L_{ab} \leq V_a - V_b \leq U_{ab}$, put directed edges $b \rightarrow a$ of length U_{ab} and $a \rightarrow b$ of length $-L_{ab}$. Let D_{ij} be the shortest-path length from i to j . The same confidence event then gives

$$\max_j \{-D_{ji}\} \leq g_i \leq \bar{g}_i := \max_j D_{ij}. \quad (26)$$

Proposition S.29 proves that these bounds are sharp in the uncertainty set defined by the supplied intervals. No new simulation, normal approximation, or additional

probability allocation is used. For a tolerance τ , let $\mathcal{C}_\tau = \{i : \bar{g}_i \leq \tau\}$. Selecting the least recorded cost within $\mathcal{C}_\tau$ preserves the simultaneous accuracy guarantee. It identifies the cheapest *certified* candidate, not necessarily the cheapest candidate whose unknown true regret is below τ .

TABLE X

LEAST ACCOUNTED COST WITH A CATALOGUE ACCURACY CERTIFICATE

Design	d	Certified choice	$\bar{g}$	Seconds	NBO (3): $\bar{g}$
Low	10	SAA (1)	0.03935	3.47	0.03883
Low	50	SAA (1)	0.03563	3.97	0.05945
Quarterly	10	Raw actor (2)	0.7543	4.85	0.1848
Quarterly	50	Vector (3)	0.7678	11.15	0.5235
Long	10	SAA (3)	0	30.42	1.984
Long	50	SAA (3)	0.4893	67.04	2.688
Intermediate	10	Vector (2)	0.8382	6.52	0.4282
Intermediate	50	NBO (3)	0.9140	12.70	0.9140

Notes: Regret bounds are in units of 10^{-4} ; the target is at most one. Parentheses identify the original stage. The catalogue is the reference plus fifteen method-stage procedures, not the full action space. Early-stage seconds are prefix allocations; stage-three seconds are complete construction/query clocks. Shared assessment and the complete comparative bill are reported separately. The selected cost is least among certified candidates only.

The fifty-dimensional Intermediate design gives a positive NBO accuracy–work result against the complete budget catalogue. NBO’s final-stage regret upper bound is 0.000091391 (rounded upward), below 10^{-4} , at mean construction and query cost 12.6993 seconds (rounded upward). It is the least-cost candidate with this certificate. The cheaper final Raw and vector actors have regret lower bounds above 0.0001874 and 0.0001064, respectively. The cheaper four-path SAA and second-stage NBO candidates remain unresolved: their regret intervals straddle 10^{-4} . Thus the certified choice does not establish actual minimum work over these unresolved alternatives.

Table X retains the choices in every other design. The final NBO candidate is within 10^{-4} of the best catalogue payoff in six cells. Both Long cells instead select

final-stage SAA; elsewhere a cheaper Raw actor, vector predictor, or low-budget SAA has the certificate. The lower-cost alternatives are part of the result, not exclusions from the comparison.

Early-stage costs are the original construction-prefix allocations, not independent clocks for a prospectively stopped procedure. Final-stage costs are measured complete three-stage construction and query processes. The shared confirmation process is charged separately and in full; adding this same bill to each candidate does not alter its ordering. The total cost of having conducted the comparison includes *all* five construction processes and that shared assessment, not just the selected candidate. Section S.24 reports these complete bills and every candidate. The closure and resulting selection are a post-freeze deterministic reanalysis of an already simultaneous family, not a newly executed stopping rule or a fresh confirmatory experiment.

10. PROSPECTIVE WORK TO AN ECONOMIC TOLERANCE

10.1. *Temporary withdrawal charges in the capital economy*

The new design keeps the original nonlinear capital production, common and idiosyncratic innovations, and terminal dispersion preference. It asks how an economist should compute current withdrawals under temporary resource charges, current utility weights, adjustment terms, and feasible caps. The future reference withdrawal is fixed at .55. Current task changes are therefore precisely changes for which the same continuation can be reused. The transport theorem characterizes additional future changes, but those perturbations are not silently added to the experimental population.

There are four periods of length $h = .25$. All integrated coefficients and sixteen stored innovation paths are exact finite-model inputs, each path having mass $1/16$. Let $\bar{y}$ denote the coordinate average and set $f(y) = \lambda \tanh\{(1 - \mu)y + \mu\bar{y}\}$, with $\lambda = .08$ and $\mu = .20$. For uniform withdrawal $a \in [.05, \bar{a}]$,

$$\begin{aligned} x(y, a) &= y + h\{c_0\mathbf{1} + f(y) - a\mathbf{1}\}, \\ Q_\theta(y, a) &= A_0\{w \log a - \eta a^2/2 - \tau a\} - W_0 a + S_0(x(y, a)). \end{aligned} \tag{27}$$

The stored A_k, W_k correspond to $\rho = .04$, and c_0 includes the original volatility correction. Subsequent transitions use the same production map, the fixed withdrawal, and the remaining stored path increments. Subsequent rewards and the terminal quadratic dispersion payoff are all included in S_0 ; the current reward in (27) is not counted twice.

The ex ante task generator uses $w \in [.8, 1.2]$, $\eta \in [.05, .15]$, $\tau \in [0, .12]$, and $\bar{a} \in [.65, .95]$, independently uniform. Initial coordinate means lie in $[-.5, .5]$ and dispersions in $[.05, .5]$. The complete generator and the two innovation arrays are recorded in the protocol. These ranges define a transparent counterfactual exercise rather than an estimated distribution of tax requests. The six query sets are fixed nested prefixes of 1,024 tasks in each dimension. A separate service is actually executed for every prefix.

The economic effect of the charge is explicit. Where the true maximizing action is interior, strict concavity implies

$$\frac{\partial a^*(\tau)}{\partial \tau} = \frac{A_0}{Q'_\theta(y, a^*(\tau))} < 0. \quad (28)$$

The constrained optimizer is nonincreasing in τ as well. The computation therefore concerns the scale of a withdrawal response with an analytically known direction, not a change of leaderboard alone. The reported zero-charge counterfactual uses the same fitted object and changes only the current charge. For a certified loss bound U_i , an external log-utility grant $g_i = \exp\{U_i / (A_0 w_i)\} - 1$ supplies utility U_i without changing production or the future policy. This is the economic normalization of the 10^{-4} tolerance; it is not an endogenous consumption-equivalent policy computed under altered state dynamics. Complete withdrawal and compensation records appear in Table S.48.

10.2. Frozen procedures, stopping, and information rights

The original implementation and finite economic inputs were frozen before the registered study. This integration recovers that exact source and executes the same fixed design. Its clock is a new machine observation, not an additional independent training population. No seed, architecture, budget, or tolerance is selected using the

integrated outcomes. The readable execution source, digest history, and complete records accompany the paper.

The seven procedures are a current-reward rule, quadratic continuation regression, radial-basis continuation regression, NBO, a shared state-and-task actor, cached sample-average optimization, and full-support enumeration. The two fitting stages use respectively 64 and 192 training states, 4 and 16 independently sampled cached paths, and 64 and 192 neural optimization epochs. Current-only and enumerated methods require one stage. Every stage is assessed on the true finite-law objective. The service stops at its first mean certified loss at most 10^{-4} ; otherwise its final outcome remains unresolved. No fallback substitutes a different method.

Quadratic, radial-basis, and NBO predictors share the sampled continuation value and derivative labels. NBO fits one scalar potential using a 32–32 tanh network plus the known terminal quadratic component. The conventional regressions use ridge-regularized value-and-derivative fitting; the radial-basis regression uses 32 centers. The shared actor receives state, weight, adjustment, charge, and cap and optimizes current plus cached future payoff without a separate scalar critic. SAA reuses its valid innovation cache and reevaluates at each queried postdecision state. The enumerated procedure uses all sixteen atoms; this structure-exploiting comparator is not withheld because it can outperform learned approximations.

The procedure’s action search and the verifier are separate. A fitted surrogate need not be concave, and its derivative-based action search is not advertised as a proof of its own global maximum. Instead, every returned feasible action is checked by (10) for the *true* objective. Section S.28.5 derives the uniform curvature bound and the outward derivative recursion, including nonlinear production and terminal dispersion. The proof covers the whole scalar interval.

A process records initialization, fitting and cache creation, all action queries, every accuracy check, and durable stage writes before reading its stop clock. Failed earlier stages remain charged. Timing metadata is measured separately and added to the service bill. Imports and other common startup are charged once to the complete comparison, not assigned fictitious per-method observations. Post-stop mechanism diagnostics do not influence the stopping rule and have their own recorded cost.

The final aggregate serialization and publication build lie outside the scientific service clock.

10.3. Complete accuracy and work results

TABLE XI

PROSPECTIVE SCALAR-INTERVAL ACCURACY

Procedure	$d = 10$ passes	$d = 50$ passes	$d = 10$ max. U	$d = 50$ max. U
Current only	0/18	0/18	4.43×10^{-4}	4.49×10^{-4}
Quadratic	18/18	18/18	4.98×10^{-6}	2.55×10^{-6}
Radial basis	18/18	18/18	5.47×10^{-7}	2.27×10^{-7}
NBO	18/18	18/18	5.77×10^{-7}	1.47×10^{-6}
Shared actor	2/18	1/18	2.60×10^{-4}	1.27×10^{-3}
Cached SAA	18/18	18/18	3.18×10^{-7}	5.80×10^{-8}
Enumerated	18/18	18/18	1.86×10^{-18}	1.41×10^{-18}

Each dimension has six separately executed query volumes and three complete streams. A pass means the mean true scalar-regret upper bound is at most 10^{-4} . Maxima retain unresolved outcomes and every declared stage.

TABLE XII

EXECUTED WORK-TO-TOLERANCE FRONTIER, DIMENSION 10

Q	Quad.	RBF	NBO	Actor	SAA	Enum.
1	0.036	0.099	0.336	$0.460^{1/3}$	0.043	0.089
4	0.059	0.056	0.176	$0.518^{1/3}$	0.055	0.075
16	0.078	0.052	0.128	$0.622^{0/3}$	0.061	0.073
64	0.098	0.155	0.207	$0.749^{0/3}$	0.118	0.124
256	0.296	0.305	0.356	$1.174^{0/3}$	0.330	0.377
1024	1.142	1.316	1.196	$2.799^{0/3}$	1.195	1.454

Mean complete warm-service seconds over three independent streams, including all fits, queries, failed checks, and durable stage records. A superscript reports fewer than three passes; such an entry is not a certified alternative for every stream. No superscript means three passes. Current-only costs and the complete stage decomposition appear in the technical record.

TABLE XIII

EXECUTED WORK-TO-TOLERANCE FRONTIER, DIMENSION 50

Q	Quad.	RBF	NBO	Actor	SAA	Enum.
1	0.045	0.059	0.100	$1.486^{1/3}$	0.050	0.049
4	0.060	0.086	0.114	$1.516^{0/3}$	0.089	0.082
16	0.117	0.115	0.168	$1.618^{0/3}$	0.121	0.161
64	0.336	0.374	0.415	$2.036^{0/3}$	0.350	0.390
256	1.337	1.449	1.415	$4.048^{0/3}$	1.401	1.689
1024	5.810	5.913	5.732	$12.720^{0/3}$	5.930	6.760

Mean complete warm-service seconds over three independent streams, including all fits, queries, failed checks, and durable stage records. A superscript reports fewer than three passes; such an entry is not a certified alternative for every stream. No superscript means three passes. Current-only costs and the complete stage decomposition appear in the technical record.

NBO attains the declared scalar-interval tolerance in 36 of its 36 complete runs, including 36 at the first declared fitting stage. Its largest returned mean-regret upper bound is 1.47×10^{-6} . The shared conditional actor attains the target in 3 of 36 runs; all remaining outcomes stay unresolved rather than being replaced. The conventional regressions and structure-exploiting simulation procedures are retained on the same accuracy and cost basis.

Across the twelve dimension–query-volume cells, the lowest mean service cost among methods certified in all three streams belongs to Quadratic in 8 cells, Cached SAA in 1 cells, Radial basis in 2 cells, NBO in 1 cells. This is an observed comparison among the declared procedures on the reported machine, not a statistical assertion of minimum possible computation. Validation work is included in every entry and becomes a substantial part of the large-query bill. Consequently, reducing the query cost alone need not produce the same ranking as reducing the total work to the economic target.

The complete comparison uses 288.476 measured seconds in its reported service environment. The sum of all method service bills is 268.750 seconds, and the separately identified post-stop mechanism calculations use 16.337 seconds. Shared startup and remaining driver and individual-record overhead are part of the full

comparison, not retrospectively assigned to a selected winner. Serialization of the final aggregate summary and publication generation are outside that clock.

The frontier is a set of observed work-to-tolerance comparisons over the registered volumes, not a claim that one neural method is universally least expensive. A conventional regression can cheaply capture this short-horizon continuation, and full-support simulation is feasible for this finite law. Their successful outcomes identify the role of reusable approximation separately from a claim about neural architecture. The stronger positive high-dimensional payoff findings in the original longer menu remain visible alongside this comparison. Different horizons, objectives, action classes, and verification obligations are not pooled into one favorable statistic.

10.4. *Symmetric prediction and decision diagnostics*

For every scalar predictor, we compute exact finite-law continuation means at common exogenous lower, middle, and upper feasible actions. We report absolute squared error, error centered across that common action set, and absolute error at the predictor’s own implemented action. We also report the true payoff loss from maximizing the predicted objective over the common three-action menu, and binary and menu misclassification frequencies. The fitting and Raw caches are regenerated across the three complete streams; stream-level records remain available. Current-only and direct actor methods do not define scalar predictors and are not assigned invented continuation risks. These diagnostics repair the action-distribution asymmetry of the earlier NBO/reference contrast assessment. They remain descriptive finite-design results over three fixed streams, not confidence intervals for every future cache or training draw. Prediction errors are evaluated against enumerated means, so a noisy squared target is not being confused with absolute risk. The connection to decisions is (4) on the common finite menu and the directly verified true-objective loss on the scalar interval. Neither connection licenses substituting a smaller average prediction risk for a protected payoff comparison between two different deployed policies.

TABLE XIV
COMMON-ACTION AND OWN-ACTION ASSESSMENT AT 1,024 QUERIES

d	Predictor	Absolute	Centered	Own-action	Menu loss
10	Quadratic	5.48×10^{-6}	1.43×10^{-7}	5.47×10^{-6}	2.17×10^{-7}
10	Radial basis	4.46×10^{-6}	1.41×10^{-8}	4.40×10^{-6}	1.79×10^{-7}
10	NBO	4.98×10^{-6}	3.24×10^{-8}	4.88×10^{-6}	4.04×10^{-7}
10	Cached SAA	4.23×10^{-6}	5.86×10^{-9}	4.21×10^{-6}	8.84×10^{-8}
10	Enumerated	0	0	0	0
50	Quadratic	2.33×10^{-6}	1.32×10^{-7}	2.30×10^{-6}	1.75×10^{-7}
50	Radial basis	1.54×10^{-6}	1.38×10^{-8}	1.49×10^{-6}	0
50	NBO	1.79×10^{-6}	9.22×10^{-8}	1.63×10^{-6}	3.01×10^{-7}
50	Cached SAA	1.30×10^{-6}	2.22×10^{-9}	1.29×10^{-6}	1.16×10^{-9}
50	Enumerated	0	0	0	0

Prediction entries are squared continuation errors relative to full-support means. Menu loss is a payoff difference on the common three-action catalogue, not scalar-interval regret. Means cover all three independently regenerated caches. Current-only and shared-actor procedures do not define scalar predictors. These are finite-design descriptive averages, not confidence intervals over unobserved training streams.

10.5. *The previous evidence and its chronology*

The previous complete experiment used higher volatility in the same capital model, sixteen declared streams per method and dimension, and a common independent final payoff assessment. The complete protocols, records, and endpoint tables remain current in technical Sections S.14 and S.15. The following facts determine how the new results are compared with that evidence.

At dimension ten, the original NBO-minus-HJB interval was approximately $[-0.000073, 0.000663]$. It did not establish superiority. After the original analytical freeze and inspection of interval widths, the common-innovation transfer was refined on the same observations. Its resulting interval was $[0.000179, 0.000411]$. At dimension fifty the original interval was $[0.001051, 0.001945]$ and the refined interval was $[0.001347, 0.001650]$. Endpoints in this paragraph are rounded outward. The refinement changes a deterministic allowance, not the simulated policies, ob-

servations, original confidence event, or earlier stopping decisions. It is labelled a post-freeze analytical reanalysis throughout the present comparison.

All four refined NBO–Raw and NBO–DPO direct intervals remained unresolved at the 10^{-4} margin. They established neither a ranking nor practical equivalence. At dimension fifty, all sixteen NBO, Raw, and DPO streams reached the original online gain target 0.0005. Their mean complete process times were approximately 357.80, 346.22, and 596.94 seconds. The specified HJB implementation reached that online target in no stream. At dimension ten no method reached the online target; the later final certificates do not change that outcome. A pooled dimension-ten HJB lower gain above 0.0005 is a method-mean statement even though no individual HJB stream had a final certificate at that target.

The original sufficient mechanism lower bounds were approximately -0.006778 and -0.004800 . Its critic–Raw continuation-risk contrasts were also unresolved. The NBO full-adapted-class regret upper bounds, rounded upward, were 0.025971 and 0.036843. These are absolute accuracy bounds with their own anchor gap, rather than a 10^{-4} near-optimality certificate. The new signed account and strengthened comparison are assessed alongside these retained outcomes.

The scalar and observation experiments provide separate information. The scalar Howard calculation reports residual, restriction, mesh, and domain accounts for its finite-grid problem. Its deployed classical policies outperform the frozen historical neural policies in the measured scalar comparisons, and the scalar NBO–DPO ranking reverses at one initial state. In the observation experiment all four protected parent comparisons have positive lower gains, while six of twelve measurement specifications retain a positive continuous-economy lower gain after sensing allowances. Each statement remains tied to its own model parameters, policy, decision grid, and information structure.

11. ECONOMIC APPLICATIONS OF THE BELLMAN STRUCTURE

The motivating economic problem combines external choices with the costly adjustment of preferences. Recursive utility changes how a given policy is evaluated; preference adjustment adds an internal choice to its Hamiltonian; and strategic interaction changes whose deviations define improvement. These distinctions de-

termine the economic interpretation of NBO. The capital economy studied above provides a setting in which policy-specific verification can be carried through to a high-dimensional diffusion. The preference, temporal-self, and game applications establish the corresponding economic objects and their own verification results.

11.1. Costly preference adjustment and portfolio choice

Let u denote risk aversion and X financial wealth. The household chooses consumption c , portfolio share p , and preference-adjustment effort θ . The preference state is reflected on a declared interval $[\underline{u}, \bar{u}] \subset (1, \infty)$, and wealth follows the portfolio diffusion:

$$\begin{aligned} du &= \theta dt + \sigma_u dW^u + dL^u - dL^{\bar{u}}, \\ dX &= \{[r + p(\mu - r)]X - c\} dt + p\sigma_X X dW^X, \quad d\langle W^u, W^X \rangle_t = \varrho dt. \end{aligned} \quad (29)$$

For a fixed consumption unit $\bar{c} > 0$, flow utility is

$$U(c, u) = \frac{1}{2}k\theta^2, \quad U(c, u) = \frac{(c/\bar{c})^{1-u}}{1-u}.$$

The objective discounts both felicity and adjustment costs at rate ρ and includes the specified wealth-exit and terminal payoff. The baseline uses $\bar{c} = 1$. Because the household controls u , the cardinal normalization across preference states is a primitive: adding a function of u changes the incentives to adjust preferences. The reflected state, positive consumption domain, and stopping data give the evaluation equation its economic and mathematical domain.

At an interior optimum with $V_X > 0$ and $V_{XX} < 0$, the three policy conditions are

$$\begin{aligned} c^* &= \bar{c}(\bar{c}V_X)^{-1/u}, & \theta^* &= V_u/k, \\ p^* &= -\frac{V_X(\mu - r)}{XV_{XX}\sigma_X^2} - \frac{\varrho\sigma_u V_{uX}}{XV_{XX}\sigma_X}. \end{aligned} \quad (30)$$

Box constraints require their corresponding feasible maximizers. The cross derivative in the portfolio rule links the external risk choice to the internal preference state. Its sign is an equilibrium object to be determined by evaluation. The adjustment

rule likewise relates effort to the shadow value of changing preferences. A higher adjustment cost lowers the value of every fixed policy and hence the optimized value; whenever an envelope derivative exists,

$$\partial_k V^k(t, u, X) = -\frac{1}{2} \mathbb{E}_{t,u,X}^{a^k} \int_t^\tau e^{-\rho(s-t)} \theta_s^2 ds \leq 0. \quad (31)$$

This comparative static holds with the feasible class and stopping payoff fixed.

The numerical application implements these choices jointly. The finite-action study evaluates complete policies by independent backward recursion. The continuous-action study trains the actor through its own payoff and verifies its hybrid policy with a complete action-box enclosure. Fresh training on nested grids separates the effect of resolution from the evaluation of a frozen coarse policy. A model-specific compensating-consumption calculation translates its cardinal utility bounds into an explicit transfer experiment. Application Sections [A.1.3](#), [A.1.3.4](#), and [A.1.8](#) give the full model, all recorded policies, refinement evidence, and compensation account.

11.2. Recursive utility and temporal selves

Recursive utility enters policy evaluation through the policy's own continuation utility. With $a = 1 - 1/\psi$, $b = 1 - \gamma$, and $bV > 0$, the Epstein–Zin aggregator used here is

$$f(c, V) = \frac{\rho}{a} \{c^a (bV)^{1-a/b} - bV\}. \quad (32)$$

The invariant utility interval and boundary payoff are part of the evaluation problem. Application Section [A.1.2](#) derives the homothetic policy conditions and verifies the matching-bequest experiment, including the barriers that keep every policy utility in the aggregator's domain. This application tests recursive evaluation while maintaining the same separation between value approximation and feasible action improvement.

Time inconsistency requires a different improvement criterion. For the continuous discount kernel

$$h(s) = \beta e^{-\rho s} + (1 - \beta) e^{-(\rho+\kappa)s}, \quad 0 < \beta \leq 1, \quad \kappa > 0,$$

two critics evaluate the same future policy at the two discount rates. Writing these values as W_1, W_2 , the current self uses $V = \beta W_1 + (1 - \beta)W_2$ and chooses an instantaneous deviation by maximizing $u(a) + \mathcal{L}^a V$. Application Section A.1.4 states the complete component equations and derives the spike-deviation condition. In the homothetic consumption economy, the equilibrium consumption share is the unique root in its admissible interval, and stronger present bias raises that share. Independently optimized finite-duration deviations check the limiting equilibrium condition. Thus the actor's objective follows from the temporal equilibrium concept, while the critics retain their policy-evaluation role.

11.3. Capital investment and strategic deviations

In the dynamic Cournot application, firm i chooses investment in its own capital, with

$$dK^i = (I^i - \delta K^i) dt + \sigma_K K^i dW^i, \quad q_i = AK^i.$$

Its flow profit is $P_N(K)q_i - I^i - b_i(I^i)^2/2$, where the market-size normalization in P_N is specified for each economy. Given rivals' policies, a critic evaluates the resulting own-firm payoff and the actor changes only its own action. Consequently, the relevant economic error is the gain from a complete unilateral deviation against the frozen rival profile.

Application Sections A.1.5 and A.1.5.3 specify the boundaries, investment sets, and market normalizations, and report independent full dynamic best responses to every principal finite neural profile. The resulting profit-unit bounds include all dates and states of the stated finite game. They also cover randomized and history-dependent unilateral deviations against those fixed Markov rivals, by finite dynamic programming. The record reports the cost of the exhaustive training oracle and any nonzero one-step equilibrium defects along with the final exploitability calculation.

These applications give the common method a precise economic interpretation. A recursive critic evaluates a utility process; preference and capital actors allocate resources and adjustment effort; a temporal actor satisfies a spike condition; and a strategic actor is assessed against unilateral best responses. Each application

supplies the boundary conditions, approximation class, and error account required by its own economic question.

12. CONCLUSION

Neural Bellman Operators organize policy evaluation, feasible improvement, and economic assessment around a specified continuation. The contribution is an economic computation with an explicit error and work account, not a new principle of optimality or a presumption that a neural approximation must dominate a conventional one.

The centered decision-loss bound identifies the errors that can change current choices. The continuation-transport theorem extends reuse to a verified neighborhood of future economic primitives and gives a condition for refreshing the evaluated object. The scalar true-objective certificate then supports actual first-success stopping with every failed check charged. These results connect approximation, economic stability, verification, and amortization without conflating their distinct assumptions.

The capital evidence contains both successful NBO decisions and useful reversals. The original high-dimensional menus establish protected payoff advantages in declared cells, while other cells favor simulation and simpler actors. The new query-volume study certifies NBO across every declared stream and retains the competitive conventional surrogates, unresolved actor outcomes, full verification bill, and own-action diagnostics. A repetition of the same frozen study is a replication of its computation, not a larger independent experiment. The finite scalar certificate does not substitute for the separate continuous-time and full-control accounts.

Recursive utility, endogenous preferences, temporal selves, and dynamic games remain part of the paper. Their utility domains, information sets, transversality conditions, and deviation criteria determine the continuations that can be evaluated and reused. The complete application models and proofs are retained. Taken together, the results specify when a learned scalar continuation supports an economically accurate decision, when transport or refresh is required, and how to compare its complete computational cost with other legitimate numerical procedures.

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